

PSP DB Infra Upgrade Runbook

Context

As part of capacity planning and to support multiple initiatives, which will introduce load to database in terms of traffic (Read-write), data size, resource utilization, we would need to scale PSP primary database vertically i.e. modify RDS instance size to next or higher instance size and storage. This document covers on planning, test and estimated extra cost.

Action Items

Prod

	Task	Steps	Status	Owner	Remarks
1	Send downtime update to stake holders via Email		DONE		
2	Create CR's for each service	PAPI: CHG1398659 RJF: CHG1398661 PSP: CHG1398660 Payroll DM: CHG1398663 Pubsub: CHG1398664 DB CR: CHG1387844	DONE		

3	Current connections status	MACHINE COUNT(1) desktop-pay-api-rollout-5cc5f54945-65fjd 5 desktop-pay-api-rollout-5cc5f54945-gtshz 5 desktop-pay-api-rollout-5cc5f54945-hm7mb 5 desktop-pay-api-rollout-5cc5f54945-pzh5w 5 desktop-pay-api-rollout-5cc5f54945-q5j6l 5 desktop-pay-api-rollout-5cc5f54945-qdth 5 desktop-pay-api-rollout-5cc5f54945-rf4kf 5 desktop-pay-api-rollout-5cc5f54945-iclvk 5 desktop-pay-api-rollout-5cc5f54945-zdwaj 5 psp-adapter-deployment-75f6b9977-5gnz4 25 psp-adapter-deployment-75f6b9977-b2rjf 25 psp-adapter-deployment-75f6b9977-cnrdt 25 psp-adapter-deployment-75f6b9977-fzfkx 25 psp-adapter-deployment-75f6b9977-gvzqk 25 psp-adapter-deployment-75f6b9977-htk8d 25 psp-adapter-deployment-75f6b9977-n94px 25 psp-adapter-deployment-75f6b9977-q75fl 25 psp-adapter-deployment-75f6b9977-sxs7c 25 psp-adapter-deployment-75f6b9977-vwdl8 25 psp-adapter-deployment-75f6b9977-vwx2z 25 psp-adapter-deployment-75f6b9977-x75h6 25 psp-adapter-deployment-7695dbdf4-bn46j 25 psp-adapter-deployment-7695dbdf4-dh7gx 25 psp-adapter-deployment-7695dbdf4-dkxm7 25 psp-adapter-deployment-7695dbdf4-fhmxm 25 psp-adapter-deployment-7695dbdf4-j8nhf 25 psp-adapter-deployment-7695dbdf4-k4sgd 25 psp-adapter-deployment-7695dbdf4-mqhh4 25 psp-adapter-deployment-7695dbdf4-ns94k 25 psp-adapter-deployment-7695dbdf4-s2np6 25 psp-adapter-deployment-7695dbdf4-t7hrh 25 psp-adapter-deployment-7695dbdf4-twjgf 25 psp-adapter-deployment-7695dbdf4-zknmw 25 psp-appd-deployment-b8c54cd6b-6mrd8 5 psp-appd-deployment-b8c54cd6b-9dmhm 5 psp-appd-deployment-b8c54cd6b-df647 5 psp-appd-deployment-b8c54cd6b-hkns7 5 psp-appd-deployment-bb88bfc4c-99t6z 5 psp-appd-deployment-bb88bfc4c-lcz5v 5 psp-appd-deployment-bb88bfc4c-s48pj 5 psp-jss-high-deployment-79694789dd-ct5hr 50 psp-jss-high-deployment-79694789dd-l8mcw 5 psp-jss-high-deployment-79694789dd-n7jch 5 psp-jss-high-deployment-cbcdcc847-6g2br 5 psp-jss-high-deployment-cbcdcc847-956fs 5 psp-jss-high-deployment-cbcdcc847-bvxx7 5 psp-jss-monitor-deployment-699fc84cc8-cj 10 psp-jss-monitor-deployment-699fc84cc8-gk 5 psp-jss-monitor-deployment-699fc84cc8-mc 5 psp-jss-monitor-deployment-98c7bc6cc-ftw 10 psp-jss-monitor-deployment-98c7bc6cc-lr6 5 psp-jss-monitor-deployment-98c7bc6cc-lxl 5 psp-jss-normal-deployment-55dd5f788f-5tm 5 psp-jss-normal-deployment-55dd5f788f-ddg 5 psp-jss-normal-deployment-55dd5f788f-wx4 5 psp-jss-normal-deployment-fc46fb9df-4zf2 18 psp-jss-normal-deployment-fc46fb9df-gjqw 15 psp-jss-normal-deployment-fc46fb9df-r54p 7 psp-qbdt-deployment-66845b697d-8l42t 15 psp-qbdt-deployment-66845b697d-8rvx6 15 psp-qbdt-deployment-66845b697d-h12pm 15 psp-qbdt-deployment-66845b697d-hzcdx 15 psp-qbdt-deployment-66845b697d-khdm5 15 psp-qbdt-deployment-66845b697d-kt9fr 15 psp-qbdt-deployment-66845b697d-mf24w 15 psp-qbdt-deployment-66845b697d-n862z 15 psp-qbdt-deployment-66845b697d-plv76 15 psp-qbdt-deployment-66845b697d-tpbds 15 psp-qbdt-deployment-85454b997c-58zsp 15 psp-qbdt-deployment-85454b997c-5kgdt 15 psp-qbdt-deployment-85454b997c-7nc8b 15 psp-qbdt-deployment-85454b997c-8hjvx 15 psp-qbdt-deployment-85454b997c-gd4mp 15 psp-qbdt-deployment-85454b997c-ich9d 15 psp-qbdt-deployment-85454b997c-p4jmd 15 psp-qbdt-deployment-85454b997c-q6fcm 15 psp-qbdt-deployment-85454b997c-qthw6 15 psp-qbdt-deployment-85454b997c-rqvbl 15 psp-qbdt-deployment-85454b997c-vrz4l 15 psp-qbdt-deployment-85454b997c-wgs4s 15 psp-sap-deployment-74d58679b8-2rdfg 5 psp-sap-deployment-74d58679b8-2xj6q 5 psp-sap-deployment-74d58679b8-6s8pc 5 psp-sap-deployment-74d58679b8-cnckw 5 psp-sap-deployment-74d58679b8-nwvp6 5 psp-sap-deployment-74d58679b8-qhbp2 5 psp-sap-deployment-74d58679b8-qr5cz 5 psp-sap-deployment-74d58679b8-w797w 5 psp-sap-deployment-b54775d48-7kz4p 5 psp-sap-deployment-b54775d48-9rfs2 5 psp-sap-deployment-b54775d48-cr427 5 psp-sap-deployment-b54775d48-gchjn 5 psp-sap-deployment-b54775d48-l97zj 5 psp-sap-deployment-b54775d48-p796w 5 psp-sap-deployment-b54775d48-sj5dx 5 psp-sap-deployment-b54775d48-z57br 5 pubsubpsp-appd-deployment-587556dd8f-6rn 50 pubsubpsp-appd-deployment-587556dd8f-cs6 8 pubsubpsp-appd-deployment-587556dd8f-kv9 50 pubsubpsp-appd-deployment-587556dd8f-xcf 50 pubsubpsp-appd-deployment-587556dd8f-z2g 6
---	----------------------------	---

6	Enable turn away page for PSP	Enable Maintenance page using the TurnAway URL	DONE		
7	Ensure all services have 0 Pods running	☒ PayrollDm ☒ RJF ☒ PAPI ☒ Pubsub ☒ PSP	DONE		
8	Monitoring of applications during change window	☒ PTC	DONE		
9	Perform RDS Infra Change	Primary DB ☒ Upgrade RDS instance size from db.r5.12xlarge to db.r5.16xlarge ☒ Increase IOPS from 40000 to 45000 ☐ Increase storage from 32 TB to 37 TB (It depends on allowed % in increase as per RDS) ☐ this will be taken care once IOPS change completes. Read Replica ☒ Increase storage from 32 TB to 37 TB (It depends on allowed % in increase as per RDS) ☒ Restart service. (not required)	DONE		
10	DB Health after change	☒ start GG extract processes ☒ EPSPWP01 ☒ EUIPWP01 ☒ DB Health check ☒ Ensure GG processes are running ☒ Ensure all changes are completed and instance is in available state via console ☒ DB Connectivity verification on Primary as well as Read Replica ☒ if required start read replica service	DONE		
11	Scale up to Desired pod counts for all services	All services should be using the new database connection. Sequence: ☒ PAPI ☒ PSP ☒ Pubsub ☒ RJF ☒ Payroll DM	DONE		

12	Ensure db connections are coming	MACHINE COUNT(1) desktop-pay-api-rollout-c7586767-25fwq 5 desktop-pay-api-rollout-c7586767-h2jdv 5 desktop-pay-api-rollout-c7586767-h6nx2 5 desktop-pay-api-rollout-c7586767-kxjfw 5 desktop-pay-api-rollout-c7586767-l4r7j 5 desktop-pay-api-rollout-c7586767-pjgps 5 desktop-pay-api-rollout-c7586767-xmj5 5 desktop-pay-api-rollout-c7586767-z58wr 5 desktop-pay-api-rollout-c7586767-zsxbi 5 ggpspuwp01.sbg-psp-prod.a.intuit.com 15 ip-172-19-0-140 1 ip-172-19-0-50 85 psp-adapter-deployment-75f6b9977-26jhj 25 psp-adapter-deployment-75f6b9977-5zf87 25 psp-adapter-deployment-75f6b9977-95zpm 25 psp-adapter-deployment-75f6b9977-h5dcc 25 psp-adapter-deployment-75f6b9977-nzqhc 25 psp-adapter-deployment-75f6b9977-sfksw 25 psp-adapter-deployment-75f6b9977-tsxxm 25 psp-adapter-deployment-75f6b9977-v9g45 25 psp-adapter-deployment-75f6b9977-w8qhv 25 psp-adapter-deployment-75f6b9977-xlm9 25 psp-adapter-deployment-7695dbdf4-22gtn 25 psp-adapter-deployment-7695dbdf4-28mr9 25 psp-adapter-deployment-7695dbdf4-2qm27 25 psp-adapter-deployment-7695dbdf4-7zbhj 25 psp-adapter-deployment-7695dbdf4-dh2lm 25 psp-adapter-deployment-7695dbdf4-fnc6l 25 psp-adapter-deployment-7695dbdf4-fwnqf 25 psp-adapter-deployment-7695dbdf4-mkqv9 25 psp-adapter-deployment-7695dbdf4-nx4fp 25 psp-adapter-deployment-7695dbdf4-p8czs 25 psp-appd-deployment-b8c54cd6b-7ws4x 5 psp-appd-deployment-b8c54cd6b-9q5sc 5 psp-appd-deployment-b8c54cd6b-g265c 5 psp-appd-deployment-bb88bfc4c-6sx28 5 psp-appd-deployment-bb88bfc4c-hhsbh 5 psp-appd-deployment-bb88bfc4c-mlsww 5 psp-jss-high-deployment-79694789dd-477dx 5 psp-jss-high-deployment-79694789dd-krjp2 5 psp-jss-high-deployment-79694789dd-tqm7v 5 psp-jss-high-deployment-cbcdcc847-9gzjt 5 psp-jss-high-deployment-cbcdcc847-qlvg8 5 psp-jss-high-deployment-cbcdcc847-sc5f5 5 psp-jss-monitor-deployment-699fc84cc8-hp 5 psp-jss-monitor-deployment-699fc84cc8-m5 5 psp-jss-monitor-deployment-699fc84cc8-qp 5 psp-jss-monitor-deployment-98c7bc6cc-g45 5 psp-jss-monitor-deployment-98c7bc6cc-q8s 5 psp-jss-monitor-deployment-98c7bc6cc-tz4 5 psp-jss-normal-deployment-55dd5f788f-nm2 5 psp-jss-normal-deployment-55dd5f788f-nx8 5 psp-jss-normal-deployment-55dd5f788f-t6l 5 psp-jss-normal-deployment-fc46fb9df-2xr2 6 psp-jss-normal-deployment-fc46fb9df-8vtv 6 psp-jss-normal-deployment-fc46fb9df-mplt 5 psp-qbdt-deployment-66845b697d-54dzc 15 psp-qbdt-deployment-66845b697d-bfq55 15 psp-qbdt-deployment-66845b697d-dg2gd 15 psp-qbdt-deployment-66845b697d-fbgrv 15 psp-qbdt-deployment-66845b697d-hf5r9 15 psp-qbdt-deployment-66845b697d-llk4hh 15 psp-qbdt-deployment-66845b697d-trnz4 15 psp-qbdt-deployment-66845b697d-vjtz4 15 psp-qbdt-deployment-66845b697d-wf6j5 15 psp-qbdt-deployment-66845b697d-xr6bp 15 psp-qbdt-deployment-85454b997c-2f8n6 15 psp-qbdt-deployment-85454b997c-4vg5d 15 psp-qbdt-deployment-85454b997c-5sj5p 15 psp-qbdt-deployment-85454b997c-7cjf7 15 psp-qbdt-deployment-85454b997c-f64r9 15 psp-qbdt-deployment-85454b997c-kl2vk 15 psp-qbdt-deployment-85454b997c-l24wx 15 psp-qbdt-deployment-85454b997c-qfm42 15 psp-qbdt-deployment-85454b997c-rhxxc 15 psp-qbdt-deployment-85454b997c-w7qtp 15 psp-sap-deployment-74d58679b8-6cmpb 5 psp-sap-deployment-74d58679b8-dflk2 5 psp-sap-deployment-74d58679b8-f6cbw 5 psp-sap-deployment-74d58679b8-j5r5t 5 psp-sap-deployment-74d58679b8-nlbnl 5 psp-sap-deployment-74d58679b8-scz5p 5 psp-sap-deployment-74d58679b8-vkrwc 5 psp-sap-deployment-74d58679b8-wwt68 5 psp-sap-deployment-b54775d48-2jlmq 5 psp-sap-deployment-b54775d48-777g5 5 psp-sap-deployment-b54775d48-9hmk9 5 psp-sap-deployment-b54775d48-dtd74 5 psp-sap-deployment-b54775d48-f999d 5 psp-sap-deployment-b54775d48-jmzqp 5 psp-sap-deployment-b54775d48-l4dv7 5 psp-sap-deployment-b54775d48-sk98g 5 pubsubpsp-appd-deployment-587556dd8f-67l 7 pubsubpsp-appd-deployment-587556dd8f-7rz 7 pubsubpsp-appd-deployment-587556dd8f-8wr 48 pubsubpsp-appd-deployment-587556dd8f-9n2 50 pubsubpsp-appd-deployment-587556dd8f-h97 50
----	----------------------------------	--

1310

15	PubSub Sanity	- Check DB connections from Pubsub - Mayank - Check for any recent message processing in splunk logs. - Check for following splunk logs in pubsub to ensure pubsub is up. <pre>(index=iks) ((sourcetype=eventrouter AND event.involvedObject.namespace=payroll-dtpayroll-pubsubpsp-usw2-prd-prd) OR (kubernetes.namespace_name::payroll-dtpayroll-pubsubpsp-usw2-prd-prd)) clustername::payroll-sgmnt-prod-usw2.cluster.k8s.local "Started application in"</pre> <pre>(index=iks) ((sourcetype=eventrouter AND event.involvedObject.namespace=payroll-dtpayroll-pubsubpsp-usw2-prd-prd) OR (kubernetes.namespace_name::payroll-dtpayroll-pubsubpsp-usw2-prd-prd)) clustername::payroll-sgmnt-prod-usw2.cluster.k8s.local "Successfully connected to"</pre>	DONE		
16	RJF, PAPI, Payroll DM Sanity	QueryConsole URL : https://qc.sap.psp.intuit.com/queryengine Set C & A cookie values in browser through plugins. Try following queries. <pre>SELECT source_company_id, fed_tax_id FROM psp_company WHERE ROWNUM < 10</pre>	DONE		
17	Make sure no jobs gets sceduled in IKS1 JSS Prod for PSP.		DONE		
18	Execute the batchjobs which could not run due to downtime.	- Expected jobs to run on PSP today between 2:PM IST to 4:30PM IST EntitlementProcessor LedgerOperationsProcessor AMOMessageProcessor RetryQbdtUnprocessedRequestProcessor PSPTtoEMSBSDDataSyncProcessor EmailGatewayProcessor AssistedUsageDataSyncProcessor PISqlJobsProcessor WorkersCompProcessor RTBAutomationProcessor IamEmailAddressProcessor - Expected jobs to run on RJF today between 2:PM IST to 4:30PM IST Complia_Adjustments All above jobs are high frequent jobs and should catch up the schedule after bringing back the service	DONE		
19	Send update to stake holders once services are back up and running.		DONE		
20	Close All CR's	PAPI: RJF: PSP: Query Console: Payroll DM: Pubsub:	DONE		

Downtime Details

Application	Downtime Start (IST)	Downtime End (IST)
PSP	2:53 PM	3:56 PM
PAPI	2:49 PM	3:44 PM
RFJ	2:42 PM	4:12 PM
Pubsub	2:50 PM	4:18 PM

Payroll DM	2:41 PM	4:21 PM
------------	---------	---------